

NARESH KOTHARI

Automotive Validation Engineer | Python Test Automation

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PROFESSIONAL SUMMARY:

Automotive Validation Engineer with 3+ years of experience in ADAS and ECU testing, specializing in Python-based test automation. Skilled in pytest and CANoe for signal validation, image comparison, and automated report generation. Experienced in HIL testing, requirement analysis, and defect tracking, with a focus on improving test efficiency through scalable automation solutions.

WORK EXPERIENCE:

System Test Engineer | Magna Electronics (via Navtech Electronics)

Project 1: BMW ADCAM (Autonomous Driving Camera Module)

- Performed system-level testing and validation of ADAS functionalities using closed-loop and open-loop methodologies.
- Evaluated lateral control features such as Lane Keep Assist (LKA), Lane Change Warning (LCW), and Lane Support Assist (LSA) through simulated driving scenarios.
- Conducted HIL simulations by replaying real-world driving videos using the vehicle environment in front of the ECU camera.
- Developed and executed test cases based on system requirements and analyzed test results.
- Hands-on experience with Vector CANoe, including trace analysis, graphic panels, measurement setup, and test configuration.

Project 2: MAZDA SVS – Surround View System

- Performed system testing of ECU software functions including Dirty Lens Detection (DLD), Pedestrian Detection (PD), Parking Line Detection (PLD), Failsafe, Calibration, HMI, and overlays
- Designed and developed Python-based automation scripts for CANoe test execution, including signal validation, image capture, and automated PDF report generation.
- Conducted KPI validation for PLD using recorded video scenarios, analyzing detection accuracy, line visibility, and environmental conditions
- Performed load testing using the RRS setup to evaluate ECU performance under different conditions .
- Assisted in automating regression test scenarios using CAPL scripting in CANoe.
- Executed automated test suites using pytest and used Jenkins for scheduled automation and reporting.
- Logged, tracked, and reported defects using Windchill, ensuring timely resolution and quality compliance.
- Hands-on experience with Vector CANoe, including IG block configuration, trace analysis, graphical panels, measurement setup, and signal monitoring.
- Collaborated with cross-functional teams to ensure feature compliance, quality validation, and on-time delivery.
- Developed a custom Python library to integrate CANoe with Robot Framework for automated test execution and reusable keyword-based validation

SKILLS:

- **Languages:** Python, C, CAPL.
- **Tools & Platforms:** CANoe, ADTF, MITE, Automation Desk, Windchill (ALM), pytest, Git, GitHub, Jenkins (Basic CI/CD usage).
- **Testing:** Detection Testing, HMI Testing, Calibration, Diagnostics, Failsafe, Load Testing.
- **Methodologies:** STLC, SDLC, V-Model.
- **Protocols:** CAN, LIN, UART, SPI, I2C, UDS
- **Validation Methods:** HIL Testing, Regression & Functional Testing, Manual & Automation Testing.

CERTIFICATIONS:

- ISTQB® Foundation Level (CTFL) – Udemy
- The Python Bible™ – Udemy
- Bash Shell Scripting – Udemy

ACHIEVEMENTS:

- Reduced project cost by automating test cases using Python, eliminating the dependency on dual automation licenses.
- Reduced testing cycle time by 20% through optimized test procedures and workflow improvements.

EDUCATION:

B.Tech – Electronics & Communication Engineering | 2019 – 2023
KKR & KSR Institute of Technology & Sciences

- Secured First Class with Distinction
- Final Year Project: Smart Parking System – Designed and implemented a prototype to monitor parking slot availability, log vehicle entry/exit times, and generate bills based on parking duration using embedded system logic.

LANGUAGES

- English
- Marwadi
- Hindi
- Telugu

DECLARATION

I hereby declare that the above-mentioned particulars are true to the best of my knowledge.

Naresh Kothari